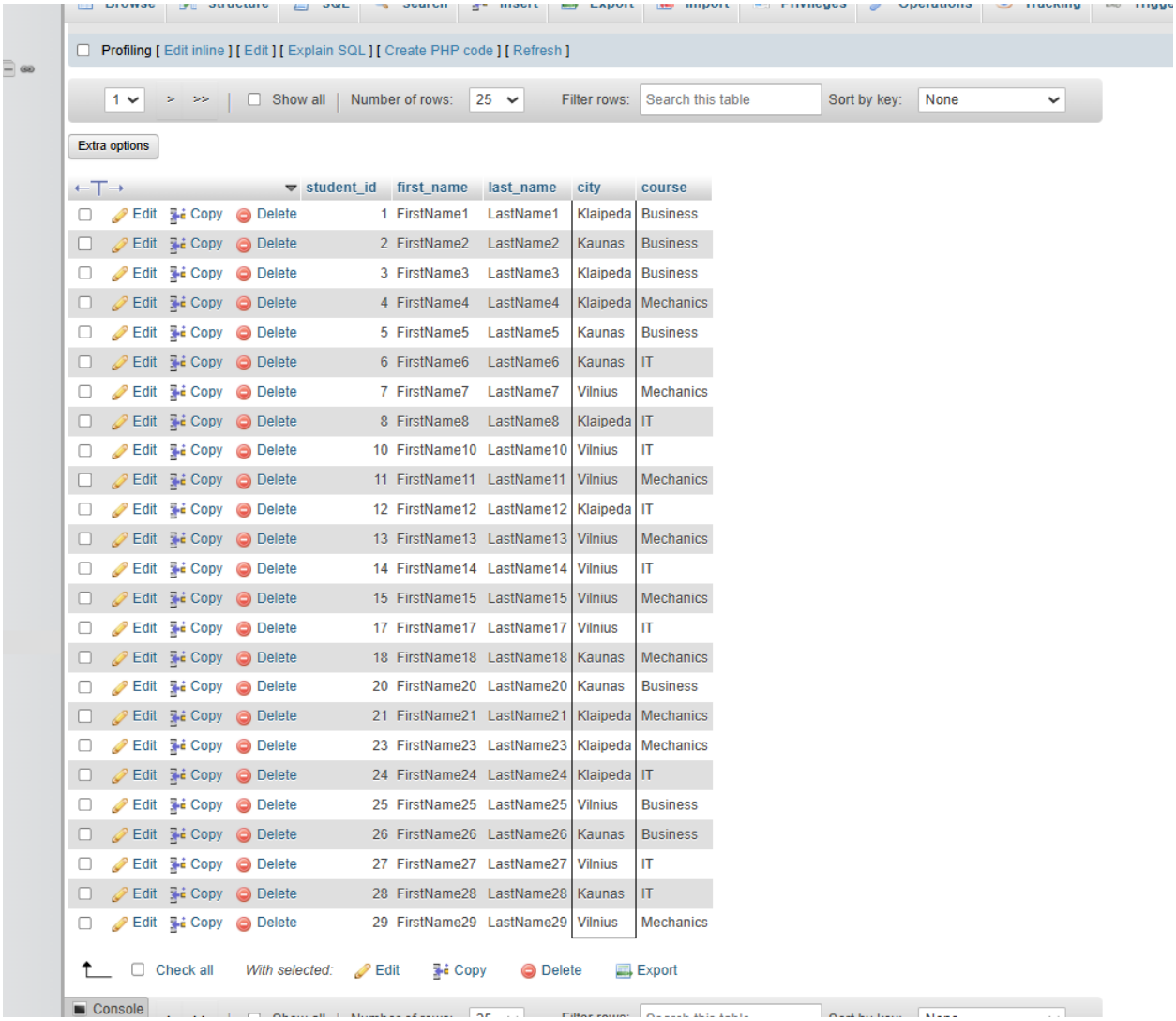
Test Report

1. Select students who are from the same city as employees working in the IT department:

```sql
SELECT *
FROM students
WHERE city IN (
    SELECT city
    FROM employees
    WHERE department = 'IT'
);
```



Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> | Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	student_id	first_name	last_name	city	course
Edit Copy Delete	1	FirstName1	LastName1	Klaipeda	Business
Edit Copy Delete	2	FirstName2	LastName2	Kaunas	Business
Edit Copy Delete	3	FirstName3	LastName3	Klaipeda	Business
Edit Copy Delete	4	FirstName4	LastName4	Klaipeda	Mechanics
Edit Copy Delete	5	FirstName5	LastName5	Kaunas	Business
Edit Copy Delete	6	FirstName6	LastName6	Kaunas	IT
Edit Copy Delete	7	FirstName7	LastName7	Vilnius	Mechanics
Edit Copy Delete	8	FirstName8	LastName8	Klaipeda	IT
Edit Copy Delete	10	FirstName10	LastName10	Vilnius	IT
Edit Copy Delete	11	FirstName11	LastName11	Vilnius	Mechanics
Edit Copy Delete	12	FirstName12	LastName12	Klaipeda	IT
Edit Copy Delete	13	FirstName13	LastName13	Vilnius	Mechanics
Edit Copy Delete	14	FirstName14	LastName14	Vilnius	IT
Edit Copy Delete	15	FirstName15	LastName15	Vilnius	Mechanics
Edit Copy Delete	17	FirstName17	LastName17	Vilnius	IT
Edit Copy Delete	18	FirstName18	LastName18	Kaunas	Mechanics
Edit Copy Delete	20	FirstName20	LastName20	Kaunas	Business
Edit Copy Delete	21	FirstName21	LastName21	Klaipeda	Mechanics
Edit Copy Delete	23	FirstName23	LastName23	Klaipeda	Mechanics
Edit Copy Delete	24	FirstName24	LastName24	Klaipeda	IT
Edit Copy Delete	25	FirstName25	LastName25	Vilnius	Business
Edit Copy Delete	26	FirstName26	LastName26	Kaunas	Business
Edit Copy Delete	27	FirstName27	LastName27	Vilnius	IT
Edit Copy Delete	28	FirstName28	LastName28	Kaunas	IT
Edit Copy Delete	29	FirstName29	LastName29	Vilnius	Mechanics

Check all With selected: Edit Copy Delete Export

Console

2. Find clients who have placed orders for products that have the same price as products ordered by other clients from 'Vilnius':

Run SQL query/queries on table students.students:

```sql
SELECT *
FROM clients
WHERE client_id IN (
    SELECT o.client_id
    FROM orders o
    JOIN products p ON o.product_id = p.product_id
    WHERE p.price IN (
        SELECT p2.price
        FROM orders o2
        JOIN products p2 ON o2.product_id = p2.product_id
        JOIN clients c2 ON o2.client_id = c2.client_id
        WHERE c2.city = 'Vilnius'
    )
);
```

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	client_id	first_name	last_name	city
Edit Copy Delete	201	Client1	LastName1	Vilnius
Edit Copy Delete	202	Client2	LastName2	Klaipeda
Edit Copy Delete	204	Client4	LastName4	Vilnius
Edit Copy Delete	212	Client12	LastName12	Siauliai
Edit Copy Delete	213	Client13	LastName13	Kaunas
Edit Copy Delete	217	Client17	LastName17	Siauliai
Edit Copy Delete	218	Client18	LastName18	Vilnius
Edit Copy Delete	230	Client30	LastName30	Siauliai
Edit Copy Delete	232	Client32	LastName32	Vilnius
Edit Copy Delete	233	Client33	LastName33	Siauliai
Edit Copy Delete	235	Client35	LastName35	Kaunas
Edit Copy Delete	237	Client37	LastName37	Klaipeda
Edit Copy Delete	238	Client38	LastName38	Klaipeda
Edit Copy Delete	242	Client42	LastName42	Kaunas
Edit Copy Delete	244	Client44	LastName44	Vilnius
Edit Copy Delete	245	Client45	LastName45	Siauliai
Edit Copy Delete	247	Client47	LastName47	Kaunas
Edit Copy Delete	249	Client49	LastName49	Vilnius

Check all With selected: Edit Copy Delete Export

3. Get all students enrolled in courses that at least one client from 'Klaipeda' has taken:

```sql
SELECT *
FROM students
WHERE course IN (
    SELECT DISTINCT course
    FROM students
    WHERE city = 'Klaipeda'
);
```

	student_id	first_name	last_name	city	course
Edit Copy Delete	1	FirstName1	LastName1	Klaipeda	Business
Edit Copy Delete	2	FirstName2	LastName2	Kaunas	Business
Edit Copy Delete	3	FirstName3	LastName3	Klaipeda	Business
Edit Copy Delete	4	FirstName4	LastName4	Klaipeda	Mechanics
Edit Copy Delete	5	FirstName5	LastName5	Kaunas	Business
Edit Copy Delete	6	FirstName6	LastName6	Kaunas	IT
Edit Copy Delete	7	FirstName7	LastName7	Vilnius	Mechanics
Edit Copy Delete	8	FirstName8	LastName8	Klaipeda	IT
Edit Copy Delete	9	FirstName9	LastName9	Panevezys	Mechanics
Edit Copy Delete	10	FirstName10	LastName10	Vilnius	IT
Edit Copy Delete	11	FirstName11	LastName11	Vilnius	Mechanics
Edit Copy Delete	12	FirstName12	LastName12	Klaipeda	IT
Edit Copy Delete	13	FirstName13	LastName13	Vilnius	Mechanics
Edit Copy Delete	14	FirstName14	LastName14	Vilnius	IT
Edit Copy Delete	15	FirstName15	LastName15	Vilnius	Mechanics
Edit Copy Delete	16	FirstName16	LastName16	Panevezys	IT
Edit Copy Delete	17	FirstName17	LastName17	Vilnius	IT
Edit Copy Delete	18	FirstName18	LastName18	Kaunas	Mechanics
Edit Copy Delete	19	FirstName19	LastName19	Panevezys	Business
Edit Copy Delete	20	FirstName20	LastName20	Kaunas	Business
Edit Copy Delete	21	FirstName21	LastName21	Klaipeda	Mechanics
Edit Copy Delete	22	FirstName22	LastName22	Panevezys	Mechanics
Edit Copy Delete	23	FirstName23	LastName23	Klaipeda	Mechanics
Edit Copy Delete	24	FirstName24	LastName24	Klaipeda	IT
Edit Copy Delete	25	FirstName25	LastName25	Vilnius	Business

Check all *With selected:* Edit Copy Delete Export

4. Find orders that were placed for products by clients who live in the same city as employees working in the 'Finance' or 'HR' departments:

Run SQL query/queries on table students.students:

```sql
SELECT *
FROM orders
WHERE client_id IN (
    SELECT c.client_id
    FROM clients c
    WHERE c.city IN (
        SELECT city
        FROM employees
        WHERE department IN ('Finance', 'HR')
    )
);
```

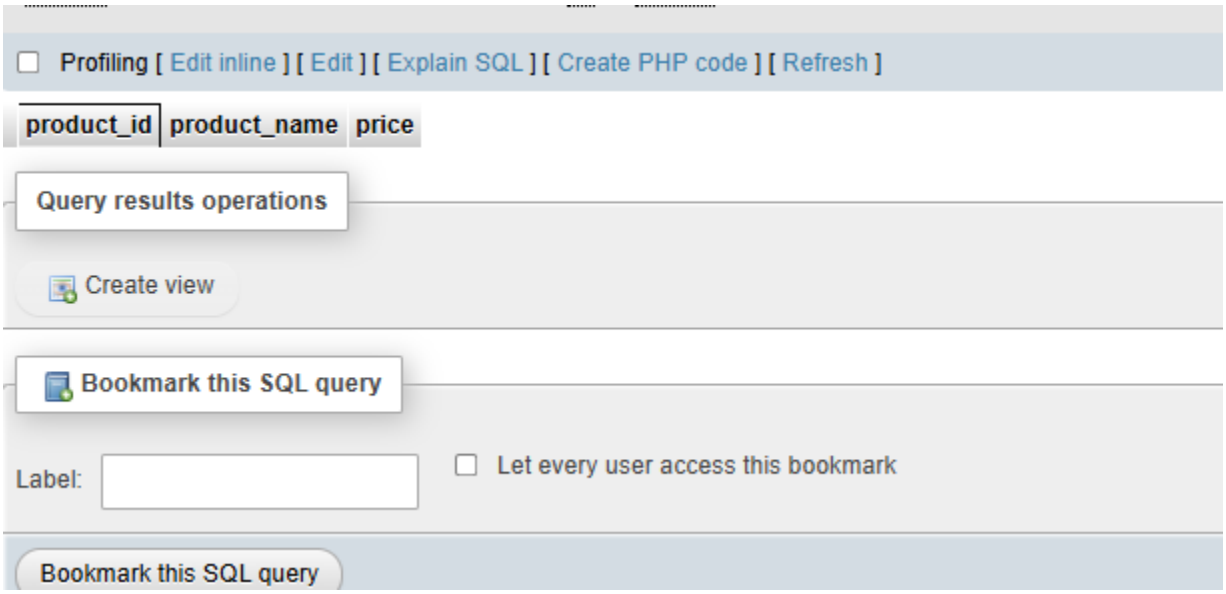
Extra options

	order_id	client_id	product_id	order_date
Edit Copy Delete	2	226	26	2024-06-24
Edit Copy Delete	3	219	46	2024-03-26
Edit Copy Delete	26	234	20	2024-06-22
Edit Copy Delete	33	238	17	2024-03-10
Edit Copy Delete	35	237	39	2024-05-01
Edit Copy Delete	36	202	5	2024-03-18
Edit Copy Delete	39	237	45	2024-03-01
Edit Copy Delete	41	223	3	2024-06-08
Edit Copy Delete	44	238	24	2024-01-11
Edit Copy Delete	49	202	24	2024-02-16
Edit Copy Delete	6	212	25	2024-05-18
Edit Copy Delete	9	230	12	2024-03-14
Edit Copy Delete	10	217	23	2024-02-18
Edit Copy Delete	18	211	41	2024-03-23
Edit Copy Delete	19	243	45	2024-06-04
Edit Copy Delete	20	233	35	2024-03-31
Edit Copy Delete	23	233	18	2024-05-23
Edit Copy Delete	25	206	11	2024-04-28
Edit Copy Delete	28	233	25	2024-03-28
Edit Copy Delete	30	217	21	2024-01-16
Edit Copy Delete	31	209	32	2024-04-27
Edit Copy Delete	45	245	13	2024-04-21
Edit Copy Delete	46	217	41	2024-01-14
Edit Copy Delete	1	225	32	2024-05-03
Edit Copy Delete	8	242	21	2024-04-11

Check all With selected: Edit Copy Delete Export

5. Select products purchased by clients who have placed orders for more than three products:

```sql
SELECT * FROM products
WHERE product_id IN (
    SELECT product_id FROM orders
    WHERE client_id IN (
        SELECT client_id FROM orders
        GROUP BY client_id
        HAVING COUNT(product_id) > 3
    )
);
```



6. Find students enrolled in the same course as employees who work in IT or Mechanics:

```sql
SELECT *
FROM students
WHERE course IN (
    SELECT department
    FROM employees
    WHERE department IN ('IT', 'Mechanics')
);
```

Extra options

	student_id	first_name	last_name	city	course
Edit Copy Delete	6	FirstName6	LastName6	Kaunas	IT
Edit Copy Delete	8	FirstName8	LastName8	Klaipeda	IT
Edit Copy Delete	10	FirstName10	LastName10	Vilnius	IT
Edit Copy Delete	12	FirstName12	LastName12	Klaipeda	IT
Edit Copy Delete	14	FirstName14	LastName14	Vilnius	IT
Edit Copy Delete	16	FirstName16	LastName16	Panevezys	IT
Edit Copy Delete	17	FirstName17	LastName17	Vilnius	IT
Edit Copy Delete	24	FirstName24	LastName24	Klaipeda	IT
Edit Copy Delete	27	FirstName27	LastName27	Vilnius	IT
Edit Copy Delete	28	FirstName28	LastName28	Kaunas	IT
Edit Copy Delete	31	FirstName31	LastName31	Vilnius	IT
Edit Copy Delete	33	FirstName33	LastName33	Kaunas	IT
Edit Copy Delete	39	FirstName39	LastName39	Klaipeda	IT
Edit Copy Delete	45	FirstName45	LastName45	Panevezys	IT
Edit Copy Delete	46	FirstName46	LastName46	Panevezys	IT
Edit Copy Delete	47	FirstName47	LastName47	Vilnius	IT
Edit Copy Delete	49	FirstName49	LastName49	Panevezys	IT

Check all *With selected:* Edit Copy Delete Export

7. Select clients who have ordered products that are priced within the same range as the products ordered by clients from 'Siauliai':

```sql
SELECT * FROM clients
WHERE client_id IN (
    SELECT client_id FROM orders
    JOIN products ON orders.product_id = products.product_id
    WHERE products.price BETWEEN
        (SELECT MIN(products.price) FROM products
         JOIN orders ON products.product_id = orders.product_id
         JOIN clients ON orders.client_id = clients.client_id
         WHERE clients.city = 'Siauliai')
        AND
        (SELECT MAX(products.price) FROM products
         JOIN orders ON products.product_id = orders.product_id
         JOIN clients ON orders.client_id = clients.client_id
         WHERE clients.city = 'Siauliai')
);
```

SELECT * | SELECT | INSERT | UPDATE | DELETE | Clear | Format | Get auto-sav

	client_id	first_name	last_name	city
Edit Copy Delete	201	Client1	LastName1	Vilnius
Edit Copy Delete	202	Client2	LastName2	Klaipeda
Edit Copy Delete	203	Client3	LastName3	Kaunas
Edit Copy Delete	204	Client4	LastName4	Vilnius
Edit Copy Delete	206	Client6	LastName6	Siauliai
Edit Copy Delete	209	Client9	LastName9	Siauliai
Edit Copy Delete	211	Client11	LastName11	Siauliai
Edit Copy Delete	212	Client12	LastName12	Siauliai
Edit Copy Delete	213	Client13	LastName13	Kaunas
Edit Copy Delete	217	Client17	LastName17	Siauliai
Edit Copy Delete	218	Client18	LastName18	Vilnius
Edit Copy Delete	219	Client19	LastName19	Klaipeda
Edit Copy Delete	221	Client21	LastName21	Kaunas
Edit Copy Delete	223	Client23	LastName23	Klaipeda
Edit Copy Delete	225	Client25	LastName25	Kaunas
Edit Copy Delete	226	Client26	LastName26	Klaipeda
Edit Copy Delete	228	Client28	LastName28	Kaunas
Edit Copy Delete	230	Client30	LastName30	Siauliai
Edit Copy Delete	232	Client32	LastName32	Vilnius
Edit Copy Delete	233	Client33	LastName33	Siauliai
Edit Copy Delete	234	Client34	LastName34	Klaipeda
Edit Copy Delete	235	Client35	LastName35	Kaunas
Edit Copy Delete	237	Client37	LastName37	Klaipeda
Edit Copy Delete	238	Client38	LastName38	Klaipeda
Edit Copy Delete	242	Client42	LastName42	Kaunas

Check all *With selected:* Edit Copy Delete Export

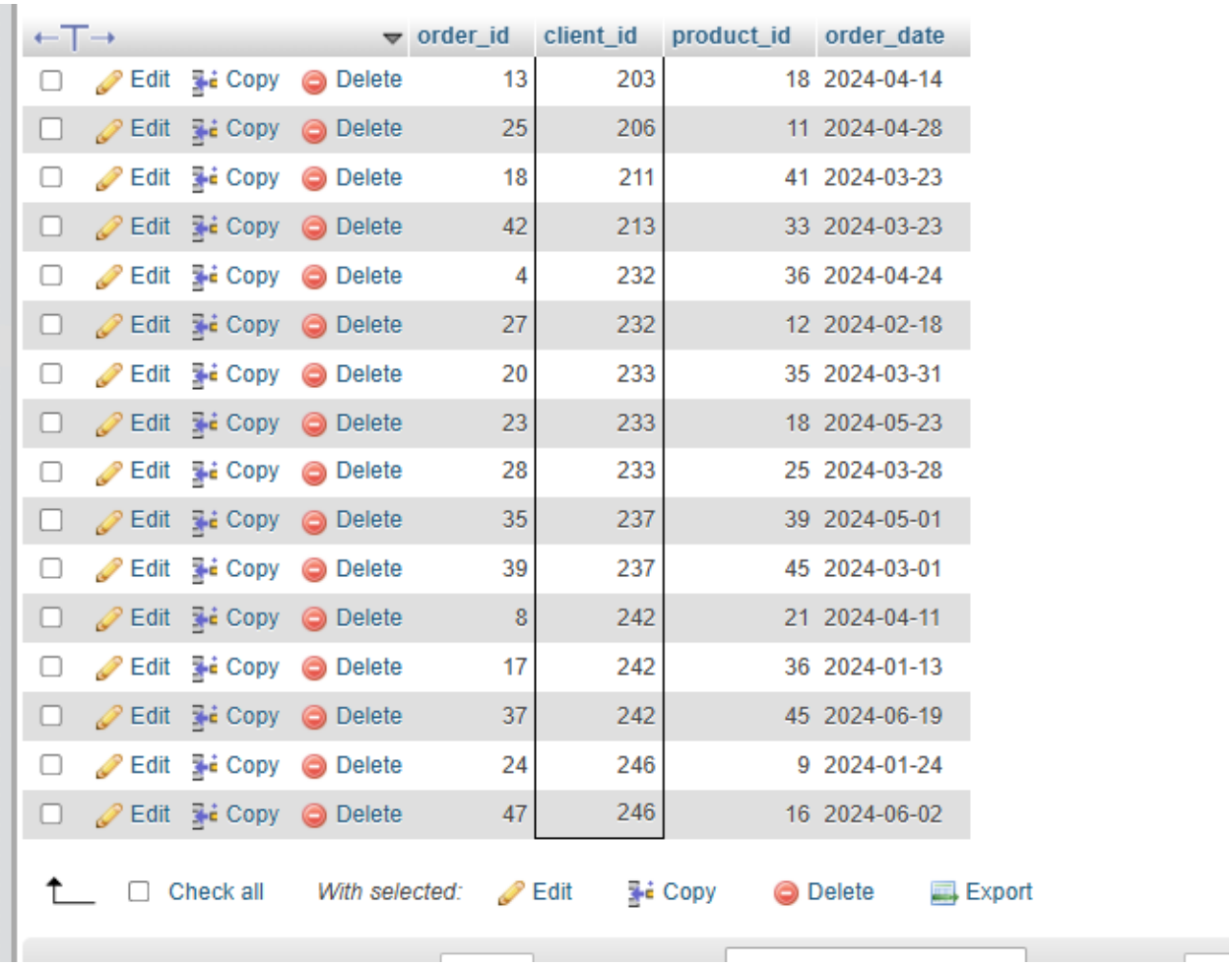
8. Get employees who live in the same city as clients who ordered products priced more than 1500:

```sql
SELECT * FROM employees
WHERE city IN (
    SELECT city FROM clients
    WHERE client_id IN (
        SELECT client_id FROM orders
        WHERE product_id IN (
            SELECT product_id FROM products WHERE price > 1500
        )
    )
);
```

	employee_id	first_name	last_name	city	department
Edit Copy Delete	101	FirstName1	LastName1	Klaipeda	HR
Edit Copy Delete	102	FirstName2	LastName2	Kaunas	IT
Edit Copy Delete	103	FirstName3	LastName3	Klaipeda	Finance
Edit Copy Delete	104	FirstName4	LastName4	Siauliai	HR
Edit Copy Delete	105	FirstName5	LastName5	Siauliai	IT
Edit Copy Delete	106	FirstName6	LastName6	Klaipeda	Finance
Edit Copy Delete	107	FirstName7	LastName7	Kaunas	IT
Edit Copy Delete	108	FirstName8	LastName8	Kaunas	Finance
Edit Copy Delete	109	FirstName9	LastName9	Siauliai	HR
Edit Copy Delete	110	FirstName10	LastName10	Vilnius	HR
Edit Copy Delete	111	FirstName11	LastName11	Klaipeda	Finance
Edit Copy Delete	112	FirstName12	LastName12	Vilnius	IT
Edit Copy Delete	113	FirstName13	LastName13	Kaunas	Finance
Edit Copy Delete	114	FirstName14	LastName14	Vilnius	HR
Edit Copy Delete	115	FirstName15	LastName15	Kaunas	HR
Edit Copy Delete	116	FirstName16	LastName16	Klaipeda	IT
Edit Copy Delete	117	FirstName17	LastName17	Siauliai	IT
Edit Copy Delete	118	FirstName18	LastName18	Vilnius	HR
Edit Copy Delete	119	FirstName19	LastName19	Vilnius	HR
Edit Copy Delete	120	FirstName20	LastName20	Vilnius	Finance
Edit Copy Delete	121	FirstName21	LastName21	Klaipeda	HR
Edit Copy Delete	122	FirstName22	LastName22	Siauliai	HR
Edit Copy Delete	123	FirstName23	LastName23	Vilnius	HR
Edit Copy Delete	124	FirstName24	LastName24	Klaipeda	HR
Edit Copy Delete	125	FirstName25	LastName25	Vilnius	HR

9. Find orders for products ordered by clients whose last name matches that of employees working in the Finance department:

```sql
SELECT * FROM orders
WHERE client_id IN (
    SELECT client_id FROM clients
    WHERE last_name IN (
        SELECT last_name FROM employees WHERE department = 'Finance'
    )
);
```



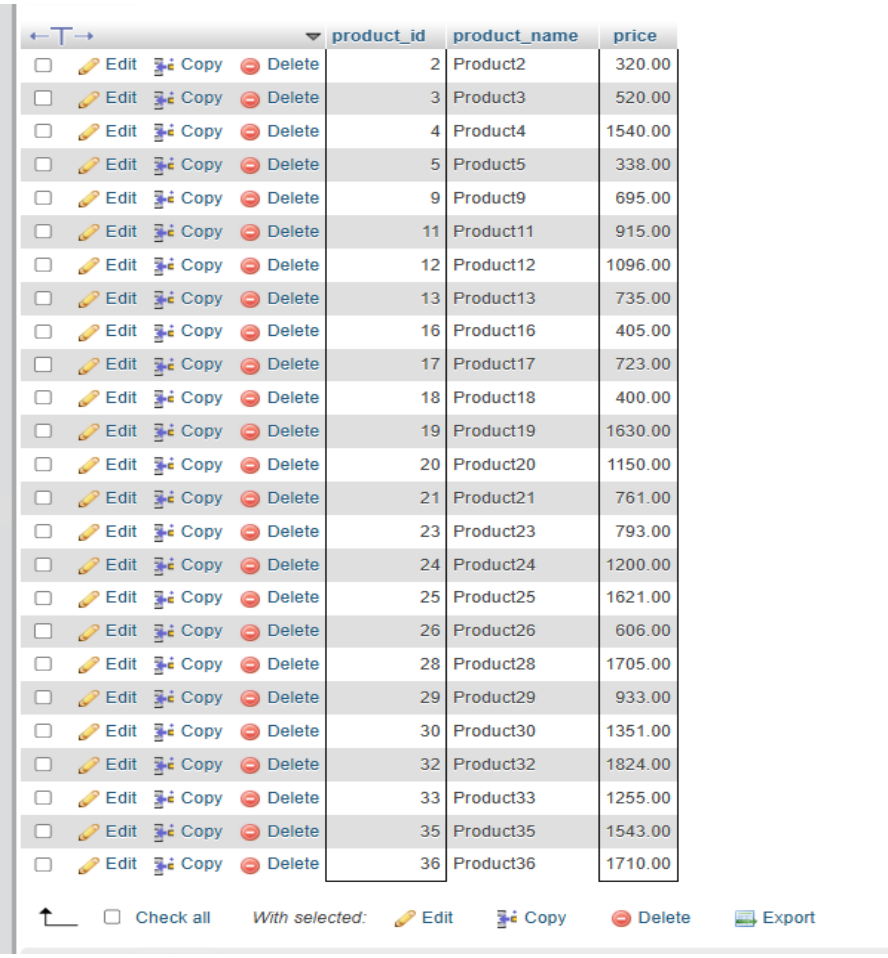
	order_id	client_id	product_id	order_date
Edit Copy Delete	13	203	18	2024-04-14
Edit Copy Delete	25	206	11	2024-04-28
Edit Copy Delete	18	211	41	2024-03-23
Edit Copy Delete	42	213	33	2024-03-23
Edit Copy Delete	4	232	36	2024-04-24
Edit Copy Delete	27	232	12	2024-02-18
Edit Copy Delete	20	233	35	2024-03-31
Edit Copy Delete	23	233	18	2024-05-23
Edit Copy Delete	28	233	25	2024-03-28
Edit Copy Delete	35	237	39	2024-05-01
Edit Copy Delete	39	237	45	2024-03-01
Edit Copy Delete	8	242	21	2024-04-11
Edit Copy Delete	17	242	36	2024-01-13
Edit Copy Delete	37	242	45	2024-06-19
Edit Copy Delete	24	246	9	2024-01-24
Edit Copy Delete	47	246	16	2024-06-02

Check all *With selected:* Edit Copy Delete Export

10. Select all products that were ordered by clients who have placed orders for products in the same price range as those ordered by employees

Run SQL query/queries on table students.orders:

```sql
SELECT *
FROM products
WHERE product_id IN (
    SELECT o.product_id
    FROM orders o
    JOIN products p ON o.product_id = p.product_id
    WHERE p.price IN (
        SELECT p2.price
        FROM products p2
    )
);
```



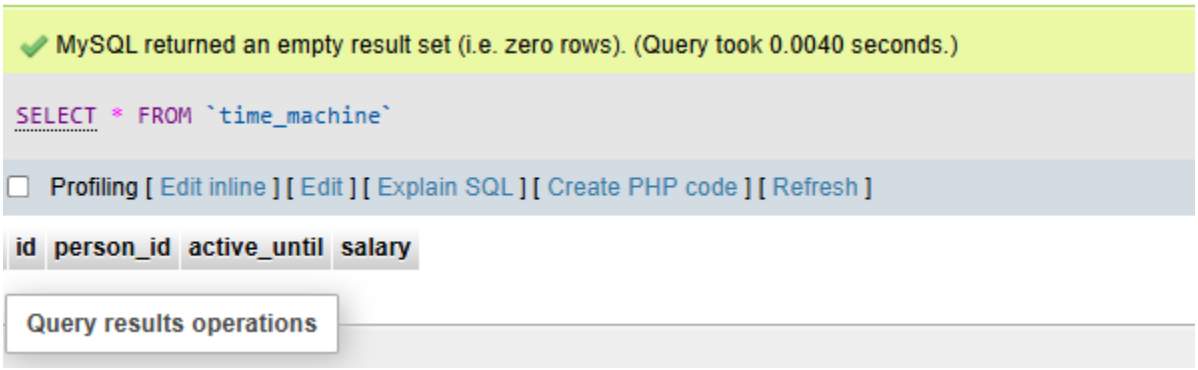
	product_id	product_name	price
Edit Copy Delete	2	Product2	320.00
Edit Copy Delete	3	Product3	520.00
Edit Copy Delete	4	Product4	1540.00
Edit Copy Delete	5	Product5	338.00
Edit Copy Delete	9	Product9	695.00
Edit Copy Delete	11	Product11	915.00
Edit Copy Delete	12	Product12	1096.00
Edit Copy Delete	13	Product13	735.00
Edit Copy Delete	16	Product16	405.00
Edit Copy Delete	17	Product17	723.00
Edit Copy Delete	18	Product18	400.00
Edit Copy Delete	19	Product19	1630.00
Edit Copy Delete	20	Product20	1150.00
Edit Copy Delete	21	Product21	761.00
Edit Copy Delete	23	Product23	793.00
Edit Copy Delete	24	Product24	1200.00
Edit Copy Delete	25	Product25	1621.00
Edit Copy Delete	26	Product26	606.00
Edit Copy Delete	28	Product28	1705.00
Edit Copy Delete	29	Product29	933.00
Edit Copy Delete	30	Product30	1351.00
Edit Copy Delete	32	Product32	1824.00
Edit Copy Delete	33	Product33	1255.00
Edit Copy Delete	35	Product35	1543.00
Edit Copy Delete	36	Product36	1710.00

Check all With selected: Edit Copy Delete Export

11(S). write an SQL query against the given time_machine file that would output active records for the specified date. Use the data from the submitted file: create a table and summarize the records. Test: with 3 dates 09/15/2023, 07/15/2023, 10/17/2023

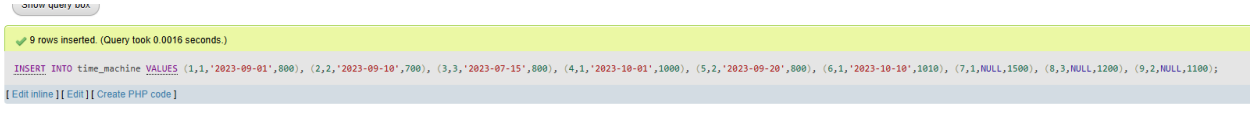
a)Create table

```sql
CREATE TABLE time_machine(
    id INT,
    person_id INT,
    active_until DATE,
    salary INT
);
```



b)Insert records

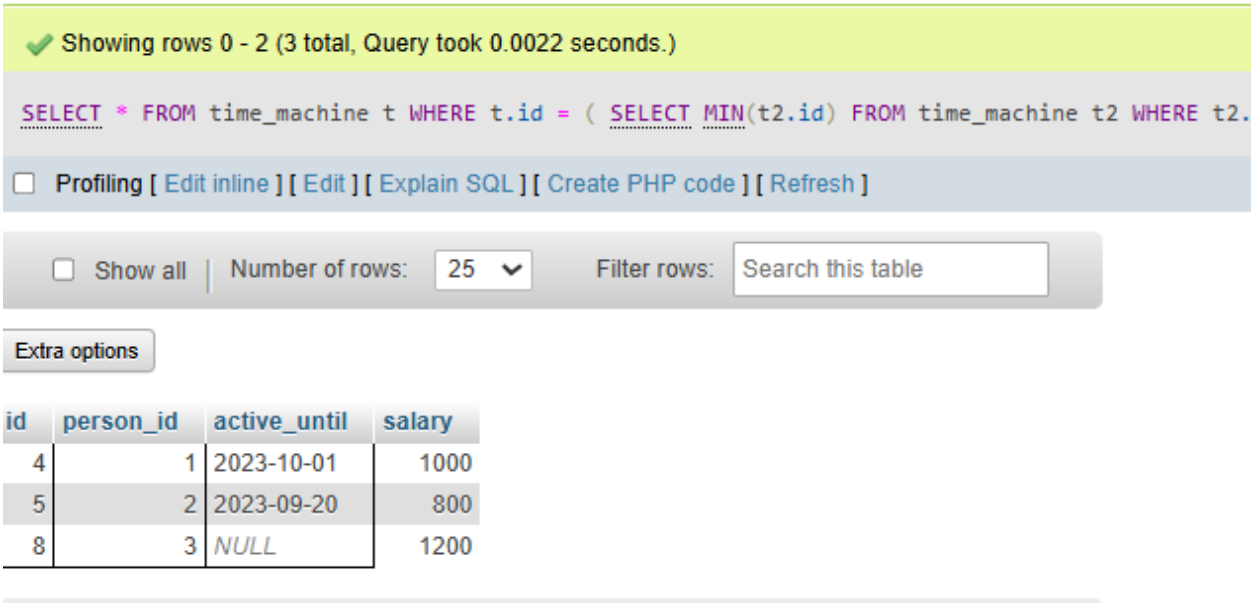
```sql
INSERT INTO time_machine VALUES
(1,1,'2023-09-01',800),
(2,2,'2023-09-10',700),
(3,3,'2023-07-15',800),
(4,1,'2023-10-01',1000),
(5,2,'2023-09-20',800),
(6,1,'2023-10-10',1010),
(7,1,NULL,1500),
(8,3,NULL,1200),
(9,2,NULL,1100);
```



c) Query for active records on a chosen date

Test date: 2023-09-15

```sql
SELECT *
FROM time_machine t
WHERE t.id =
(
    SELECT MIN(t2.id)
    FROM time_machine t2
    WHERE t2.person_id = t.person_id
      AND (t2.active_until IS NULL
           OR t2.active_until >= '2023-09-15')
);
```



id	person_id	active_until	salary
4	1	2023-10-01	1000
5	2	2023-09-20	800
8	3	*NULL*	1200

Test date: 2023-07-15

```sql
SELECT *
FROM time_machine
WHERE active_until IS NULL
   OR active_until >= '2023-07-15';
```

id	person_id	active_until	salary
1	1	2023-09-01	800
2	2	2023-09-10	700
3	3	2023-07-15	800
4	1	2023-10-01	1000
5	2	2023-09-20	800
6	1	2023-10-10	1010
7	1	*NULL*	1500
8	3	*NULL*	1200
9	2	*NULL*	1100

Test date: 2023-10-17

```sql
SELECT *
FROM time_machine
WHERE active_until IS NULL
   OR active_until >= '2023-10-17';
```

